

MTH-TOP, Topology

ATONU ROY CHOWDHURY, achowdhu@ictp.it

Contents

1 21 September 2026

2

General Information About the Course

- **Instructor:** Daniele Zuddas

Email: dzuddas@units.it

UniTS Website: <https://www.units.it/persona/index.php/from/abook/persona/33957>

§1 21 September 2026

Roughly speaking, a topological space is a set of *points* together with a notion of which points are “close” to a given one, i.e. a system of neighbourhoods of each point. To specify this, we declare which subsets are *open*.

Definition 1.1 (Topology). Let X be a set and let $\mathcal{P}(X)$ denote its power set, i.e. the set of all subsets of X . A **topology** on X is a collection $\mathcal{T} \subseteq \mathcal{P}(X)$ of subsets of X such that:

(T1) $\emptyset \in \mathcal{T}$;

(T2) $X \in \mathcal{T}$;

(T3) $\mathcal{T}$ is closed under arbitrary unions: if $\{U_\alpha\}_{\alpha \in A}$ is any collection of elements of $\mathcal{T}$ (the index set A may be finite or infinite), then

$$\bigcup_{\alpha \in A} U_\alpha \in \mathcal{T}; \quad (1.1)$$

(T4) $\mathcal{T}$ is closed under pairwise intersections: if $U, V \in \mathcal{T}$, then $U \cap V \in \mathcal{T}$.

The pair $(X, \mathcal{T})$ is called a **topological space**.

Property (T3) allows arbitrary families, while (T4) only concerns pairs. Nevertheless, (T4) implies the corresponding property for finite families, by induction.

Proposition 1.1

Property (T4) is equivalent to:

(T4') for every $n \in \mathbb{N}$ and every $U_1, \dots, U_n \in \mathcal{T}$,

$$\bigcap_{i=1}^n U_i \in \mathcal{T}. \quad (1.2)$$

Proof. (T4') implies (T4) by taking $n = 2$. The converse follows by induction on n , using

$$\bigcap_{i=1}^{n+1} U_i = \left(\bigcap_{i=1}^n U_i \right) \cap U_{n+1}. \quad (1.3)$$

■

When the topology is clear from the context (or not specified), one simply speaks of “the topological space X ”.

Remark 1.1 (Subsets and characteristic functions). There is a natural bijection between subsets of X and functions from X to $\{0, 1\}$:

$$\mathcal{P}(X) \longleftrightarrow \{\psi : X \rightarrow \{0, 1\}\}, \quad A \mapsto \chi_A, \quad \psi \mapsto \psi^{-1}(1), \quad (1.4)$$

where χ_A is the characteristic function of A . One checks easily that these two maps are inverse to each other.

Definition 1.2. Let $(X, \mathcal{T})$ be a topological space. A subset $U \subseteq X$ is **open** if $U \in \mathcal{T}$. A subset $C \subseteq X$ is **closed** if its complement $X \setminus C$ is open.

So closed sets are exactly the complements of open sets. We will use De Morgan's laws: for any family $\{A_\alpha\}_{\alpha \in A}$ of subsets of X ,

$$X \setminus \bigcup_{\alpha} A_{\alpha} = \bigcap_{\alpha} (X \setminus A_{\alpha}), \quad (1.5)$$

$$X \setminus \bigcap_{\alpha} A_{\alpha} = \bigcup_{\alpha} (X \setminus A_{\alpha}). \quad (1.6)$$

The two laws are dual to each other, and proving them is a basic exercise in set theory. Using them, the axioms (T1)–(T4) can be dualized to properties of closed sets.

Proposition 1.2 (Properties of closed sets)

Let X be a topological space.

- (C1) $\emptyset$ and X are closed.
- (C2) An arbitrary intersection of closed sets is closed.
- (C3) A finite union of closed sets is closed.

Proof. (C1) The complement of X in X is $\emptyset$, which is open; the complement of $\emptyset$ is X , which is open. Hence both X and $\emptyset$ are closed. In particular $\emptyset$ and X are both open and closed (“clopen”).

(C2) Let $\{C_\alpha\}$ be a family of closed sets. To check that $\bigcap_{\alpha} C_{\alpha}$ is closed, we must check that its complement is open. By De Morgan,

$$X \setminus \bigcap_{\alpha} C_{\alpha} = \bigcup_{\alpha} (X \setminus C_{\alpha}). \quad (1.7)$$

Each $X \setminus C_{\alpha}$ is open by hypothesis, so the union is open by (T3).

(C3) is proved in the same way, using the other De Morgan law and (T4'). ■

Equivalently, one can *define* a topology by declaring the closed sets, as long as they satisfy (C1)–(C3): the open sets are then the complements of the closed sets.

The following examples are defined on every set X .

Example 1.1 (Trivial/indiscrete topology)

$\mathcal{T}_{\text{triv}} = \{\emptyset, X\}$. The only open sets are $\emptyset$ and X , and these are also the only closed sets.

Example 1.2 (Discrete topology)

$\mathcal{T}_{\text{disc}} = \mathcal{P}(X)$. Every subset is open, and therefore every subset is also closed. Axioms (T1)–(T4) hold trivially since all subsets are allowed.

Example 1.3 (Cofinite topology)

$$\mathcal{T}_{\text{cof}} = \{\emptyset\} \cup \{U \subseteq X : X \setminus U \text{ is finite}\}. \quad (1.8)$$

The empty set has to be included by hand in the definition (if X is infinite, $X \setminus \emptyset = X$ is not finite). The open sets are thus the empty set and all subsets with finite complement.

It is not entirely obvious that this is a topology. It is more convenient to check the properties of closed sets. The closed sets of the cofinite topology are exactly X and the finite subsets of X . Then:

- X is closed by definition, and $\emptyset$ is closed because it is finite (it has 0 elements);
- an arbitrary intersection of closed sets is closed: if one of the sets is finite, the intersection is contained in it and hence finite; otherwise all the sets equal X and so does the intersection;
- a union of two closed sets is closed: it is either X (if one of the two is X) or a union of two finite sets, which is finite.

By [Proposition 1.2](#) and the remark above, $\mathcal{T}_{\text{cof}}$ is a topology.

Proposition 1.3

A topological space X is discrete if and only if all its singletons $\{x\}$ are open.

Proof. If X is discrete, every subset is open, in particular every singleton. Conversely, suppose all singletons are open. The empty set is open by (T1), and every non-empty subset $A \subseteq X$ is the union $A = \bigcup_{x \in A} \{x\}$ of open sets, hence open by (T3). ■

Proposition 1.4

If X is finite, then the cofinite topology on X equals the discrete topology on X .

Sketch (exercise). If X is finite, then every subset of X has a finite complement, so every subset is open in $\mathcal{T}_{\text{cof}}$. (If X is infinite, the two topologies differ: a singleton has infinite complement, so it is not cofinite-open.) ■

To construct more interesting topologies, we need the notion of a base. In general, the collection of all open sets is very large and hard to describe explicitly. A base is a smaller collection that can be described in a simple way and that determines the topology.

Definition 1.3 (Base). Let $(X, \mathcal{T})$ be a topological space. A collection $\mathcal{B} \subseteq \mathcal{T}$ of *open* sets is a *base* for $\mathcal{T}$ if every open set of X is a union of elements of $\mathcal{B}$.

It is important that $\mathcal{B}$ consists of open sets, as we'll see now.

Proposition 1.5 (Characterization of open sets via a base)

Let $(X, \mathcal{T})$ be a topological space and let $\mathcal{B} \subseteq \mathcal{T}$ be a collection of open sets. Then $\mathcal{B}$ is a base for $\mathcal{T}$ if and only if for every subset $U \subseteq X$ the following holds:

$$U \in \mathcal{T} \iff \forall x \in U \exists B_x \in \mathcal{B} \text{ such that } x \in B_x \subseteq U. \quad (1.9)$$

Proof. Assume that $\mathcal{B}$ is a base. Suppose that for every $x \in U$ there is $B_x \in \mathcal{B}$ with $x \in B_x \subseteq U$. Then $U = \bigcup_{x \in U} B_x$. Each B_x is open, so U is a union of open sets and is therefore open.

Conversely, suppose U is open. Since $\mathcal{B}$ is a base, $U = \bigcup_{i \in I} B_i$ for some $B_i \in \mathcal{B}$. If $x \in U$, then $x \in B_i$ for some i , and $B_i \subseteq U$; so $B_x := B_i$ works.

The converse (the property implies that $\mathcal{B}$ is a base) follows by the same argument. For every open U and every $x \in U$, there is $B_x \in \mathcal{B}$ with $x \in B_x \subseteq U$. Then $U = \bigcup_{x \in U} B_x$, is an union of the basic open sets. Every open set is a union of elements of $\mathcal{B}$, so $\mathcal{B}$ is a base. ■

Example 1.4

For every set X , the collection of all singletons $\{\{x\} : x \in X\}$ is a base for the discrete topology: singletons are open in the discrete topology, and every subset is a union of singletons (Proposition 1.3).

The following theorem tells us when a collection of subsets of a set (with no topology given yet) is a base for some topology.

Theorem 1.6

Let X be a set and let $\mathcal{B} \subseteq \mathcal{P}(X)$ be a collection of subsets of X . Then $\mathcal{B}$ is a base for some topology on X if and only if the following two properties hold:

- (B1) $X = \bigcup_{B \in \mathcal{B}} B$;
- (B2) for every $B_1, B_2 \in \mathcal{B}$ and every $x \in B_1 \cap B_2$, there exists $B_3 \in \mathcal{B}$ such that $x \in B_3 \subseteq B_1 \cap B_2$.

Moreover, in this case the topology is unique: there is exactly one topology on X having $\mathcal{B}$ as a base.

This is very useful: we do not need to describe the (large) collection of all open sets, only a simple collection $\mathcal{B}$ satisfying (B1) and (B2), and this determines a unique topology.

Proof. ($\Rightarrow$) Suppose $\mathcal{B}$ is a base for a topology $\mathcal{T}$. Then X is open, hence a union of elements of $\mathcal{B}$; since every element of $\mathcal{B}$ is a subset of X , this union is X , which gives (B1). If $B_1, B_2 \in \mathcal{B}$, they are open, so $B_1 \cap B_2$ is open, and for $x \in B_1 \cap B_2$ Proposition 1.5 gives $B_3 \in \mathcal{B}$ with $x \in B_3 \subseteq B_1 \cap B_2$, which is (B2).

($\Leftarrow$) Assume (B1) and (B2). Define

$$\mathcal{T}_{\mathcal{B}} := \left\{ \bigcup_{j \in J} B_j : J \text{ arbitrary index set, } B_j \in \mathcal{B} \right\}, \quad (1.10)$$

that is, the collection of all possible unions of elements of $\mathcal{B}$. We must show that $\mathcal{T}_{\mathcal{B}}$ is a topology.

First, we record a description of $\mathcal{T}_{\mathcal{B}}$ analogous to Proposition 1.5. Note that at this point X is just a set, so we cannot use that proposition; we prove directly that for $U \subseteq X$,

$$U \in \mathcal{T}_{\mathcal{B}} \iff \forall x \in U \exists B_x \in \mathcal{B} : x \in B_x \subseteq U. \quad (1.11)$$

If $U \in \mathcal{T}_{\mathcal{B}}$, then U is a union of elements of $\mathcal{B}$, so each $x \in U$ lies in one of them, and that one is contained in U . Conversely, if the right-hand side holds, then $U = \bigcup_{x \in U} B_x \in \mathcal{T}_{\mathcal{B}}$.

Now we check the axioms.

- (T1): $\emptyset$ is the empty union, so $\emptyset \in \mathcal{T}_{\mathcal{B}}$.
- (T2): $X \in \mathcal{T}_{\mathcal{B}}$ by (B1).
- (T3): a union of unions of elements of $\mathcal{B}$ is again a union of elements of $\mathcal{B}$.
- (T4): let $U, V \in \mathcal{T}_{\mathcal{B}}$ and $x \in U \cap V$. By (1.11) there are $B, B' \in \mathcal{B}$ with $x \in B \subseteq U$ and $x \in B' \subseteq V$. By (B2) there is $B'' \in \mathcal{B}$ with

$$x \in B'' \subseteq B \cap B' \subseteq U \cap V. \quad (1.12)$$

Hence $U \cap V$ satisfies the right-hand side of (1.11), so $U \cap V \in \mathcal{T}_{\mathcal{B}}$.

So $\mathcal{T}_{\mathcal{B}}$ is a topology. Every element of $\mathcal{B}$ belongs to $\mathcal{T}_{\mathcal{B}}$ (a union with a single term), and by definition every element of $\mathcal{T}_{\mathcal{B}}$ is a union of elements of $\mathcal{B}$; thus $\mathcal{B}$ is a base for $\mathcal{T}_{\mathcal{B}}$.

Uniqueness. Let $\mathcal{T}'$ be any topology on X for which $\mathcal{B}$ is a base. Every $U \in \mathcal{T}'$ is a union of elements of $\mathcal{B}$, so $U \in \mathcal{T}_{\mathcal{B}}$. Conversely, every element of $\mathcal{T}_{\mathcal{B}}$ is a union of elements of $\mathcal{B} \subseteq \mathcal{T}'$, hence belongs to $\mathcal{T}'$ by (T3). Thus $\mathcal{T}' = \mathcal{T}_{\mathcal{B}}$. ■

Now, we'll see some examples of topologies on $\mathbb{R}$.

Example 1.5 (The Euclidean topology). Let

$$\mathcal{B}_E = \{(a, b) : a, b \in \mathbb{R}, a < b\} \quad (1.13)$$

be the collection of all open bounded intervals in $\mathbb{R}$. (Here (a, b) denotes the open interval $\{x \in \mathbb{R} : a < x < b\}$; the notation $]a, b[$ is sometimes used to avoid confusion with ordered pairs.)

We check the hypotheses of [Theorem 1.6](#). Property (B1) holds because $\mathbb{R} = \bigcup_{n \in \mathbb{N}} (-n, n)$. For (B2), let $(a, b), (c, d) \in \mathcal{B}_E$ and $x \in (a, b) \cap (c, d)$. Then

$$x \in (\max\{a, c\}, \min\{b, d\}) = (a, b) \cap (c, d), \quad (1.14)$$

and the right-hand side is again an open bounded interval (it is non-empty, since it contains x).

Hence $\mathcal{B}_E$ is a base for a unique topology on $\mathbb{R}$, called the *Euclidean topology* (also the *standard topology*). This is the topology used in calculus and analysis, and it is the default topology on $\mathbb{R}$: when we speak of $\mathbb{R}$ as a topological space without specifying the topology, we mean this one. The elements of $\mathcal{B}_E$ are called *basic open sets*.

- Open intervals (a, b) are open, being basic open sets.
- The rays $(a, +\infty)$ and $(-\infty, b)$ are open, since

$$(a, +\infty) = \bigcup_{n \in \mathbb{N}} (a, a + n), \quad (-\infty, b) = \bigcup_{n \in \mathbb{N}} (b - n, b), \quad (1.15)$$

are unions of basic open sets.

- Singletons are closed: the complement of $\{a\}$ is

$$\mathbb{R} \setminus \{a\} = (-\infty, a) \cup (a, +\infty), \quad (1.16)$$

a union of two open sets.

- Closed intervals $[a, b]$ are closed, since $\mathbb{R} \setminus [a, b] = (-\infty, a) \cup (b, +\infty)$ is open.
- Half-open intervals such as $[a, b)$ are in general neither open nor closed in the Euclidean topology.

Example 1.6 (The Sorgenfrey line). Now consider a different collection:

$$\mathcal{B}_S = \{[a, b) : a, b \in \mathbb{R}, a < b\}. \quad (1.17)$$

Proposition 1.7

$\mathcal{B}_S$ is a base for a topology on $\mathbb{R}$, called the *Sorgenfrey topology* (the *Sorgenfrey line*, or the topology of right half-open intervals), denoted $\mathbb{R}_S$.

Proof. (B1) holds because $\mathbb{R} = \bigcup_{n \in \mathbb{N}} [-n, n]$. For (B2), the intersection of two such intervals is $[a, b] \cap [c, d] = [\max\{a, c\}, \min\{b, d\}]$; if it contains a point x , it is itself a non-empty interval of the same form, so it belongs to $\mathcal{B}_S$, and we can take B_3 equal to the intersection. ■

Comparison with the Euclidean topology. Let $a < b$. For every c with $a < c < b$, the interval $[c, b]$ is a basic open set of the Sorgenfrey topology, and

$$(a, b) = \bigcup_{a < c < b} [c, b]. \quad (1.18)$$

Therefore every basic open set of the Euclidean topology is open in the Sorgenfrey line, and consequently

$$\mathcal{T}_E \subseteq \mathcal{T}_S. \quad (1.19)$$

The two topologies are *not* the same: the interval $[a, b]$ is open in the Sorgenfrey line but not open in the Euclidean topology (no open interval (c, d) with $c < a < d$ is contained in $[a, b]$). So the Sorgenfrey topology has strictly more open sets than the Euclidean topology, and consequently it also has more closed sets: every Euclidean closed set is closed in $\mathbb{R}_S$.

Some properties of the Sorgenfrey line:

- $[a, +\infty)$ is open, because $[a, +\infty) = \bigcup_{n \in \mathbb{N}} [a, a + n]$ is a union of basic open sets.
- $[a, b]$ is both open and closed (clopen). It is open by definition, and its complement

$$\mathbb{R} \setminus [a, b] = (-\infty, a) \cup [b, +\infty) \quad (1.20)$$

is open, since $(-\infty, a) = \bigcup_{n \in \mathbb{N}} [a - n, a)$ and $[b, +\infty)$ is open as above.

- Closed intervals $[a, b]$ and singletons are closed (they are Euclidean closed), as are all Euclidean closed sets.
- The interval $(a, b]$ is not open in the Sorgenfrey line: the point b belongs to it, but there is no basic set $[b, c)$, $c > b$, contained in $(a, b]$.